

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The penetration of artificial intelligence (AI) tools into higher education has transformed how an academic student searches for information, plans assignments, drafts academic writing assignments, and manages academic learning tasks. In the current study, the use of an AI is investigated as an independent variable which describes the frequency, purposes, and depth that tertiary students make of various AI tools, such as ChatGPT and Gemini, in their academic work and critical thinking disposition is conceived as a mediating variable explaining how the use of AI translates into results in terms of academic productivity. Within the Malaysian context, the proposal highlights how little empirical work has been done to explore perspectives on the application of AI from students, in terms of how it affects dispositions in critical thinking and the level of productivity in their academic work.

Recent scholarship has had a mixed attitude towards the generative AI (GenAI) - both an opportunity for people (e.g., faster access to explanations, feedback, and examples) and a risk (e.g., overreliance, dilution of cognitive effort, and concerns over integrity). Systematic reviews of ChatGPT in education highlight this duality: ChatGPT is used by students as a learning assistant and writing aid, but responsibly integrating ChatGPT means training, governance, and explicit guidelines to avoid misuse and overreliance (Albadarin et al., 2024; Mai et al., 2024; Abdallah et al., 2025). This chapter brings together empirical and theoretical literature from recent research (2020+) to develop a coherent rationale of what we are going to study in this chapter: air usage - academic productivity relationships, air usage - developing critical thinking disposition

relationships, and developing critical thinking disposition - academic productivity relationships of tertiary students in Malaysia.

2.2 Artificial intelligence usage in contemporary higher education

Schemes of AI in higher education refer more and more to GenAI systems that can produce coherent text, support conversational responses and facilitate ideation, summarization, translation, and explanation. Across empirical studies reviewed in recent syntheses, students tend to report that they are using ChatGPT to clarify concepts, obtain examples, brainstorm essay structure and revise their writing (functions which could allow for faster workflow and save time spent on routine academic processes Albadarin et al. 2024; Abdallah et al. 2025). These patterns are consistent with the proposition in this proposal that AI tools are frequently employed as learning aids and in the preparation of assignments and learning materials and the main interest is in understanding how behaviors in use are related to students' level of productivity and their engagement in higher-order thinking.

However, the "performance promise" of AI does not necessarily translate to actual gains in learning. Meta-analytic evidence suggests that ChatGPT-supported learning interventions, on average, can increase learning performance and moderately increase higher-order thinking; importantly, this does not occur in a 'one-size-fits-all' manner, as learning intervention effects vary by their learning models, durations and how ChatGPT is used (e.g. tutor vs. tool vs. partner), suggesting that the outcomes of such interventions are not uniform and are influenced by pedagogy and the cognitive engagement of the learner (Wang & Fan, 2025). A similar systematic review and meta-analysis of experimental studies similarly finds ChatGPT to be capable of enhancing academic performance and higher-order thinking propensities while lowering mental effort, again

recognizing the importance of design choices, and that "less effort" can indicate either greater efficiency, or less cognitive work, depending on context (Deng et al., 2025).

This tension is central to the current study: academic productivity is not viewed as simply "doing more," but as efficiently and effectively accomplishing academic tasks while producing quality learning outputs. If AI decreases the mental work without enhancing the ability to reflect evaluation of material presented, students certainly will be able to finish work quicker but will be less engaged in analytic and evaluative thinking that is important for strong learning and learning of student. On the other hand, if the use of AI is integrated into practices of active learning and self-regulation, then AI may be supporting deeper exploration and reflective learning.

2.3 Theoretical foundations supporting the study variables

2.3.1 Technology acceptance perspectives and AI usage

Because the use of AI in a student context is optional in many environments, an explanation of adoption is often offered using the lens of technology acceptance. In the proposal, the theoretical framework combines the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) to explain the reasons behind the use of AI tools by students, coupled with theories that address learning to explain how its use may affect thinking and productivity. While TAM focuses on perceived usefulness and perceived ease of use as determinants of technology use (Davis, 1989), UTAUT goes further in that performance expectancy, effort expectancy, social influence, and facilitating conditions all play a role (Venkatesh et al., 2003).

Recent empirical work applying UTAUT to ChatGPT adoption depicts how these constructs work in a higher education context. A cross-country study comparing university students from the UK

and Nepal identified performance expectancy, effort expectancy, and social influence had a significant impact on adoption intention in both countries but that anxiety operated differently across countries (Budhathoki et al., 2024). This type of evidence supports the logic of the proposal that students are much more likely to use AI tools when they believe there will be tangible academic benefits (such as higher efficiency, better outcomes of their tasks), find the tools easy to use, and see others (peers) using them. While acceptance theories explain the use of AI, they do not suffice to explain what occurs cognitively when AI use becomes standard-either cognitively, or via the need to incorporate critical thinking and self regulated learning perspectives.

2.3.2 Critical-thinking disposition as a cognitive–motivational mechanism

Critical thinking is usually distinguished into (a) skills (analysis, inference, evaluation) and (b) disposition, or the steadfast willingness or inclination to use those skills. The proposal defines critical thinking disposition as a tendency to use analytical, reflective and evaluative thinking when dealing with information, including AI generated content. This "willingness to think critically" is crucial in AI contexts because the quality of the AI output is variable and often requires verification, triangulation and judgement of relevance and appropriateness.

Empirical work in higher education continues to demonstrate that critical thinking is related to academic outcomes in meaningful ways. For example, research of first-year university students has found positive relationships between critical thinking and academic performance, which suggests students who are able to better analyze and evaluate information are better able to perform academically (Guamanga et al., 2024). Meta-analytic evidence also supports an overall positive relation between critical thinking and academic achievement regardless of the educational context (Orhan, 2022). Together, these findings add to an argument for addressing critical-thinking disposition as an underlying mechanism for how AI use may affect academic productivity,

particularly in instances where AI use involves new demands on evaluation and judgement instead of basic information retrieval.

2.3.3 Self-regulated learning as the “bridge” from AI use to academic outcomes

Self-regulated learning (SRL) theory focuses on active planning, monitoring, use of strategies and reflection among learners. In GenAI-supported learning, SRL is quite relevant because students need to make decisions about when to consult AI and how to frame the prompt, as well as how to check the outputs, and how to incorporate or reject suggestions. Syntheses in GenAI education consistently come to the conclusion that positive learning impacts are more likely when the use of AI is scaffolded and integrated into active learning routines rather than being used as a shortcut (Mai et al., 2024; Deng et al., 2025; Wang and Fan, 2025). This appears consistent with the argument made in this proposal that acceptance alone is insufficient; one must be cognitively involved and there must be some self-regulation for the outcomes to be meaningful.

2.4 AI usage among tertiary students

In this study, use of AI represents the frequency and purpose of student use (i.e., drafting, synthesizing, generating ideas, checking grammar, solving problems; and asynchronous and synchronous) and the "depth" of use (surface-level copying vs. use of an interacting dialogue and iterative improvement). This definition is supported by findings from reviews that students use ChatGPT as a virtual assistant for on-demand explanations, writing support, and personalized learning guidance (Albadarin et al., 2024). Yet recent reviews also demonstrate that educational benefits are highly variable: in some educational contexts, ChatGPT-based use is linked with engagement and performances in learning; in others, its use may have detrimental consequences for independent learning or collaborative competencies if overused (Abdallah et al., 2025; Albadarin et al., 2024).

Moreover, recent scholarship focuses on how AI gets used rather than whether AI gets used. Meta-analyses show ChatGPT to have a power on higher order thinking moderated by the way of integration (role of ChatGPT, course type, learning model), which suggests that students cognitive orientation such as the disposition to evaluation and reflection may be the key differentiator in the outcome (Wang & Fan, 2025).

2.5 Critical-thinking disposition in AI-mediated learning

The proposal refers to critical-thinking disposition as an inclination to apply analytic, reflective, and evaluative thinking when faced with information, instead of experiencing a passive acceptance of it. In an AI-assisted learning setting, this tendency is a practical necessity: Students are required to assess accuracy, identify hallucinations or oversimplifications and determine if AI-produced explanations are suitable to the course requirements and scholarly standards. The more general GenAI literature increasingly poses effective use as including "verification and judgement" instead of passive consumptions (Abdallah et al., 2025; Abbas et al., 2024).

Evidence from samples outside strictly educational ones also helps to clarify the risk of lower cognitive effort. A large chi 2025 study surveying knowledge workers found that higher confidence in GenAI was related to less critical thinking while higher self-confidence in task ability was related to more critical thinking; qualitatively, participants reported that critical thinking shifted towards information verification and response integration (Lee et al. 2025). Although this research is not restricted to students, the mechanism directly applies: if students trust AI too readily that could engender less critical scrutiny of AI that can undermine the very dispositions for quality learning and sound academic judgement.

2.6 Academic productivity in tertiary education

The concept of academic productivity in this proposal is efficiency and effectiveness in completing academic tasks, such as time management, quality of performance, and learning outcomes. It further focuses that productivity for students is not only the amount of work completed, but what quality and level is met within required timeframes - encapsulates completion of assignments, performance and improvement levels. This definition is useful because it considers both "speed" and "quality" which is essential to consider when evaluating AI tools, which may increase the pace of output generation, but may endanger originality or depth if used without critical thinking.

Academic productivity in the recent empirical literature is often operationalized with academic performance (grades, test scores), efficiency in carrying out tasks, the feeling of effective learning, and lowering the burden of workload. Meta-analyses of educational interventions using ChatGPT report students would improve learning performance and reduce mental effort - which is a pathway to better productivity, if the time savings resulted in a better allocation of study time or deeper practice (Wang & Fan, 2025; Deng et al., 2025). However, as systematic reviews caution, lowering productivity does not always benefit, as there is a risk of misuse that can lead to independent learning as well the increasing technical stress (i.e., technostress) and academic integrity issues that compromise academic outcomes in the long run (Abdallah et al., 2025; Abbas et al., 2024).

2.7 AI usage and academic productivity

The first objective is to identify the relationship between the usage of AI and academic productivity of Malaysian university students. Across experimental and quasi-experimental research, learning supported by ChatGPT has been linked to aggregate positive learning performance outcomes. In a recent meta-analysis that synthesized 51 studies (Nov 2022 - Feb 2025), ChatGPT had a large positive effect on learning performance, and moderate positive effects on learning perception and

higher order thinking, although effects depended on the type of course, duration, learning model, and ChatGPT's role in instruction (Wang & Fan, 2025). Another systematic review and meta-analysis of experimental studies came to a similar conclusion that ChatGPT interventions can increase academic performance and higher order thinking propensities, and decrease mental effort, but with methodological limitations necessitating cautious interpretation and future rigorous research (Deng et al., 2025).

These results suggest two plausible pathways of productivity. The first one is a direct pathway of efficiency: Searching, summarizing, and drafting can be sped up by AI tools, allowing the academic tasks to be finished faster - this is in line with the framing of the proposal of a direct effect of AI usage -> academic productivity. The second is a reallocation pathway: if AI frees up time for low level activities, students may get to spend more time on higher level learning activities (practice, revision, reflection), which in turn, will improve the quality and results. Yet this pathway is conditional in that if "reduced effort" translates to "reduced thinking," productivity is sometimes higher in the short term but the quality of learning diminishes over time. This conditionality is stressed repeatedly in the systematic reviews that call for institutional guidelines, AI literacy, and pedagogical frameworks for the responsible use (Abdallah et al., 2025; Mai et al., 2024).

2.8 AI usage and critical-thinking disposition

The second objective is to investigate the impact of the use of AI in the critical thinking disposition of Malaysian university students. Recent literature suggests that, depending on the way that it is enacted, AI use can either support or undermine critical thinking. On the more positive side, meta-analytic evidence suggests that ChatGPT can promote higher-order thinking in certain situations especially when combined with scaffolds (such as structured prompts, problem-based learning, frameworks aligned to Bloom's taxonomy) hinting towards this to be possible - using AI as a

partner in exploration and reflective learning. This is in good agreement with the conceptual nature of the proposal that asserts that AI can be used to foster reflective learning if properly embedded in academic processes.

On the risk side there is increasing evidence that confidence in AI outputs can lead to reduced critical scrutiny. The study of a survey of the CHI 2025 report found that greater confidence in the GenAI was related to less critical thinking, which implies cognitive offloading in which users only trust AI, instead of actively evaluating the content (Lee et al., 2025). Complementary to this, both empirical and theoretical arguments in high education have been made that the constant or excessive use of ChatGPT for academic tasks may weaken cognitive abilities and promote "laziness" or less effort on the part of the student, especially when the student replaces AI output with their own reasoning (Abbas et al, 2024). The implication for the present research is that the use of AI is not inherently developmental for critical thinking disposition rather that it may interact with students motivation, learning strategy and institutional expectations producing positive and/or null and/or negative associations based on the patterns of use and learning design.

2.9 Critical-thinking disposition and academic productivity

The third objective is to study the relationship between the disposition of critical thinking and academic productivity between Malaysian tertiary students using AI. Beyond the domain of AI, the critical thinking literature has also consistently associated students' critical thinking capacities with academic performance indicators. For example, evidence based on the first year college students reveal positive correlations between critical thinking and academic performance, thus the view that the more able students to analyze and evaluate the information will be better at academic performance worth it (Guamanga et al., 2024). Meta-analytic statistics have shown a positive relationship between critical-thinking and academic success in general across studies, reinforcing

the idea that an orientation to critical thinking leads to more favorable academic achievement (Orhan, 2022).

When mapped into the present study's definition of academic productivity as efficiency, quality, and learning outcomes, critical-thinking disposition can be theorized to support academic productivity through a number of mechanisms. Students with proclivities for critical evaluation are more likely to recognize weak arguments, to locate inconsistencies and to revise drafts towards improved quality of output and consequently to increase productivity by reducing rework and increasing the likelihood of meeting assessment criteria. In environments rich with AI, the same disposition promotes verifying AI generated content, cross-checking sources, and integrating outputs responsibly which improves both learning outcomes as well as quality of tasks. This understanding is laid out in the premises of the proposal that cares: Students who think critically about Artificial Intelligence will be able to use it in a way that produces more productivity.

2.10 Critical-thinking disposition as a mediator between AI usage and academic productivity

The placement of critical-thinking disposition as a mediating variable in the study and explicit suggestions for direct and indirect effects are: AI Usage, in turn, would lead to academic productivity, AI Usage, in turn, would lead to critical-thinking disposition, and critical-thinking disposition, in turn, would lead to academic productivity. Mediation is theoretically possible because AI tools can alter the cognition requirements of academic work. When students use AI in the process of brainstorming, and drafting, they may either (a) accept their responses with little thinking, and thereby undercuts the enactment of critical thinking, or (b) view AI's output as something from which to start and that needs to be judged, compared in light of course materials, and adapted-clearly one of the latter approaches reinforces evaluative habits and dispositions over time. The existence of such divergent pathways is also reflected in the large body of systematic

reviews that can report on performance benefits and concerns surrounding over-reliance and undermined independent learning (Abdallah et al., 2025; Munaye et al., 2025; Abbas et al., 2024).

Recent results from a meta-analytic study further support the logic of mediation by demonstrating the moderate and mediated effects of ChatGPT on higher-order thinking (Wang & Fan, 2025). This suggests that cognitive outcomes are not only a matter of access to AI, but crucially include patterns of engagement of learners, as well as instructional framing. If critical-thinking disposition can form and be formed by the way learners engage with AI output(s) - to either verify or question or integrate the AI output(s) - then it seems reasonable to expect that students high in critical-thinking disposition will translate the use of AI into more reliable productivity gains (higher quality work within time constraints), while students low in critical-thinking disposition may opt for surface-level productivity gains combined with weaker learning quality. The empirical evidence showing suppression of critical thinking by AI confidence (Lee et al., 2025) suggests a concrete mechanism of the workings of this mediator: In the absence of a critical thinking process, the use of AI may lead to a decrease in cognitive effort and fragile learning.

2.11 Synthesis of literature and research gaps

A few patterns have emerged across recent research to justify the need for the present research. First, while aggregated evidence suggests that ChatGPT can improve learning performance, higher-order thinking, and perceived learning outcomes results vary according to pedagogy, duration, and role of AI in other words, "AI use" is not a uniform intervention (Wang & Fan, 2025; Deng et al., 2025). Second, there has been a consistent emphasis from systematic reviews pointing to a risk profile in the form of overreliance, concerns for integrity and lower levels of independent learning, which implies that cognitive dispositional governors as well as the structure of governance are central to outcomes (Abdallah et al., 2025; Mai et al., 2024; Munaye et al., 2025).

Third, the new evidence revealed that confidence in GenAI can diminish critical thinking suggests that critical thinking is not only an outcome but rather a dynamic mechanism influenced by the patterns of AI interaction (Lee et al., 2025).

Within the Malaysian context, the novel presents a lack of research in terms of exploring the interplay between the use of AI, critical thinking disposition, and academic productivity, which has been a limitation to the development of evidence-based guidelines for responsible integration of AI into a higher learning institution, as the proposed model responds to the gap by empirically testing direct and mediated pathways where critical thinking disposition lens is expected explain the reasons why the use of AI may translate into meaningful productivity for some students and not others.

2.12 Chapter summary

This chapter has reviewed contemporary literature on AI usage in higher education, emphasizing GenAI tools and the mixed evidence surrounding their academic value. Recent systematic reviews and meta-analyses provide support for potential learning and performance benefits while underscoring variability and risk. Technology acceptance theories (TAM/UTAUT) help explain why students adopt AI tools, but the literature suggests that academic outcomes depend on deeper cognitive mechanisms—particularly critical-thinking disposition and self-regulated engagement. Anchored in the proposal’s conceptual framework, the literature supports investigating critical-thinking disposition as a mediator linking AI usage to academic productivity among Malaysian tertiary students.

References

- Abbas, M., Jam, F. A., & Khan, T. I. (2024). Is it harmful or helpful? Examining the causes and consequences of generative AI usage among university students. *International Journal of Educational Technology in Higher Education*, 21(1), Article 10.
- Abdallah, N., Katmah, R., Khalaf, K., & Jelinek, H. F. (2025). Systematic review of ChatGPT in higher education: Navigating impact on learning, wellbeing, and collaboration. *Social Sciences & Humanities Open*, 12, 101866.
- Albadarin, Y., Saqr, M., Pope, N., & Tukiainen, M. (2024). A systematic literature review of empirical research on ChatGPT in education. *Discover Education*, 3, Article 60.
- Budhathoki, T., Zirar, A., Njoya, E. T., & Timsina, A. (2024). ChatGPT adoption and anxiety: A cross-country analysis utilising the unified theory of acceptance and use of technology (UTAUT). *Studies in Higher Education*, 49(5), 831–846.
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.
- Deng, R., Jiang, M., Yu, X., Lu, Y., & Liu, S. (2025). Does ChatGPT enhance student learning? A systematic review and meta-analysis of experimental studies. *Computers & Education*, 227, 105224.
- Guamanga, M. H., Saiz, C., Rivas, S. F., & Almeida, L. S. (2024). Analysis of the contribution of critical thinking and psychological well-being to academic performance. *Frontiers in Education*, 9, 1423441. <https://doi.org/10.3389/feduc.2024.1423441>

- Lee, H.-P. (H.), Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)* (Article 1121, pp. 1–22). Association for Computing Machinery.
- Mai, D. T. T., Da, C. V., & Hanh, N. V. (2024). The use of ChatGPT in teaching and learning: A systematic review through SWOT analysis approach. *Frontiers in Education*, 9.
- Munaye, Y. Y., Admass, W., Belayneh, Y., Molla, A., & Asmare, M. (2025). ChatGPT in education: A systematic review on opportunities, challenges, and future directions. *Algorithms*, 18(6), 352.
- Orhan, A. (2022). The relationship between critical thinking and academic achievement: A meta-analysis study. *Psycho-Educational Research Reviews*, 11(1), 283–299.
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.
- Wang, J., & Fan, W. (2025). The effect of ChatGPT on students' learning performance, learning perception, and higher-order thinking: Insights from a meta-analysis. *Humanities and Social Sciences Communications*, 12, Article 621.